



Image Classification with Convolutional Neural Network.

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Abstract

This project aimed to classify traffic sign images into five categories using deep learning. Traffic sign recognition is important in road and driving-related systems because signs need to be identified accurately from image data. The original dataset contained separate Train and Test folders, but the labelled Train folder was used as the main source of class labels. After checking the dataset, corrupted images were removed, and the cleaned labelled images were split inside the notebook into training, validation, and held-out tests. Several models were developed and compared, including a baseline convolutional neural network, a deeper convolutional neural network with batch normalization and dropout, optimizer variants using Adam and SGD, a no-dropout ablation model, and MobileNetV2 transfer learning with feature extraction and fine-tuning. Model comparison was based on validation performance, while the held-out test set was only used for final evaluation. The deeper CNN without dropout achieved the strongest validation performance and was selected as the final model. On the held-out test set, it achieved an accuracy of 0.9864 and a weighted F1-score of 0.9864. Overall, the custom CNN model generalized well and performed better than the MobileNetV2 models for this dataset.

1. Introduction

Traffic sign recognition is an image classification task where a model predicts the correct class of a traffic sign from an input image. This is useful in road safety, driver assistance, and autonomous driving systems because traffic signs provide important information for decision-making. However, classification can be challenging when images have different brightness, quality, size, or visual similarity between classes.

This project focuses on classifying five traffic sign categories using deep learning. A baseline CNN was first developed, followed by deeper CNN models, optimizer comparison, dropout ablation, and MobileNetV2 transfer learning. The aim was to compare these methods and identify which approach performed best. Models were compared using validation accuracy, precision, recall, F1-score, loss, and training time. The best validation model was then evaluated on a held-out test set to check generalization.

2. Dataset

The dataset used in this project contained traffic sign images arranged into Train and Test folders. The labelled Train folder was used as the main source of class labels because its images were stored inside class-named subfolders. The five classes were Cautions, Crossings, Direction, No Entry, and Speed Limit.

Before modelling, the images were checked for corrupted files. After removing corrupted images, 16,065 valid labelled images remained. These cleaned images were split inside the notebook into training, validation, and test sets. The validation set was used for model comparison and model selection, while the held-out test set was used only once for final evaluation.

Training samples: 11221
Validation samples: 2424
Test samples: 2420

Figure 1: Train, validation, and test split output showing zero group overlap.

A group-based split was used to avoid related image groups appearing in more than one split. The group overlap between training, validation, and test sets was 0, which helped reduce data leakage.

Methodology

This project used staged modeling roads to compare different deep learning methods for traffic sign classification. All images were resized and then passed into the models. For the custom CNN models, the image size was set to 96×96 , while MobileNetV2 used 224×224 . Because this is the expected input size for the pre-trained model. Pixel values were normalized, and data augmentation was applied to the training images using rotation, width shift, height shift, zoom, and brightness changes. This helped increase image variation during training.

The first model was a baseline CNN with convolutional layers, max-pooling layers, fully connected layers, and a softmax output layer for the five classes. A deeper CNN was then developed using more convolutional blocks with batch normalization. Dropout was also tested as a regularization method. To compare training behavior, the deeper CNN was trained with both Adam and SGD optimizers.

Experiments and Results

Several experiments were carried out to compare models fairly. All main model comparisons were based on validation performance, while the held-out test set was used only once for the final selected model.

Baseline CNN Performance

The baseline CNN was used as the first reference model. It achieved a validation accuracy of 0.9790 and a weighted F1-score of 0.9788, showing that a simple CNN could already learn useful visual patterns from the dataset. The classification report in Figure X shows strong performance for most classes, especially Direction and SpeedLimit. However, Crossings had the lowest recall (0.82) and F1-score (0.89), which suggests that this class was harder for the baseline model to recognize.

Baseline Classification Report:

	precision	recall	f1-score	support
Cautions	0.86	0.98	0.92	240
Crossings	0.97	0.82	0.89	209
Direction	1.00	1.00	1.00	477
No Entry	1.00	0.98	0.99	419
SpeedLimit	0.99	1.00	1.00	1079
accuracy			0.98	2424
macro avg	0.96	0.96	0.96	2424
weighted avg	0.98	0.98	0.98	2424

Figure 2: Baseline CNN validation classification report.

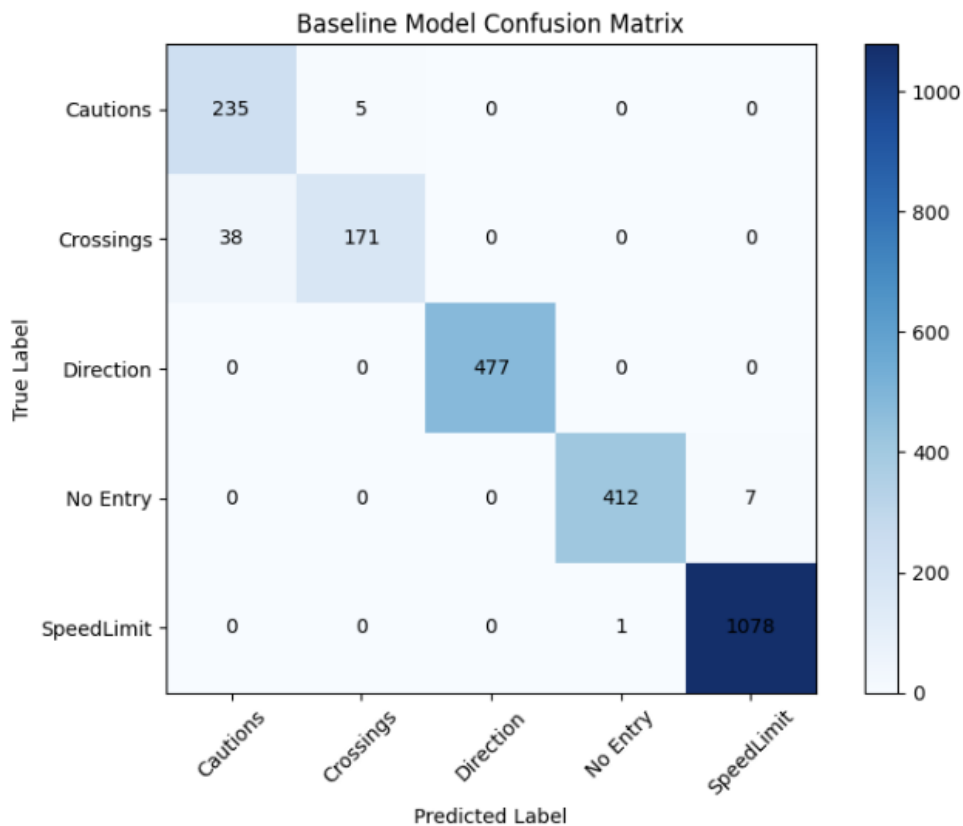


Figure 3: Baseline CNN validation confusion matrix.

The baseline confusion matrix shows that most classes were predicted correctly. The largest error was for Crossings, where some images were predicted as Cautions. This matches the classification report, where Crossings had the lowest recall. Direction and SpeedLimit were classified very strongly by the baseline model.

Baseline CNN vs Deeper CNN

The deeper CNN was compared with the baseline CNN to check whether adding more convolutional layers improved performance. The baseline CNN achieved a validation accuracy of

0.9790 and a weighted F1-score of 0.9788. The deeper CNN with dropout and Adam optimizer achieved a higher validation accuracy of 0.9843 and a weighted F1-score of 0.9843.

This shows that the deeper architecture improved performance slightly. The extra convolutional layers and batch normalization helped the model learn stronger image features. However, this improvement came with higher training time. The baseline CNN took 882.57 seconds, while the deeper CNN took 1508.79 seconds.

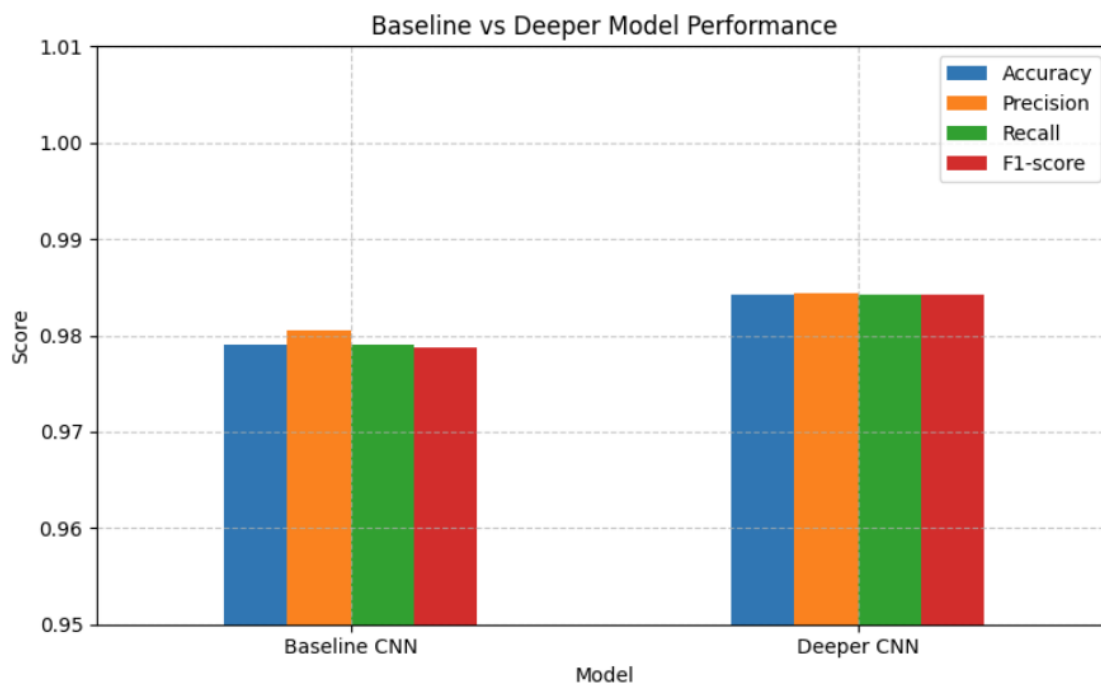


Figure 4: Validation performance comparison between the baseline CNN and deeper CNN.

Computational Efficiency

The models had different training times depending on their architecture, optimizer, and input image size. The baseline CNN was the fastest model, taking 882.57 seconds. The deeper CNN with Adam took 1508.79 seconds, showing that adding more layers increased training time. The deeper CNN

with SGD took 3091.07 seconds, which was the longest training time. This shows that SGD gave stronger validation performance, but it required more computation.

The MobileNetV2 models used larger 224 * 224 input images, while the custom CNN models used 96 * 96 images. This increased the processing cost for transfer learning. However, the MobileNetV2 models still trained faster than the SGD and no-dropout deeper CNN models. The experiments were run in Kaggle with GPU acceleration, which helped reduce training time compared with CPU-only training.

Training time comparison of the evaluated models:

Model	Training Time (seconds)
Baseline CNN	882.57
Deeper CNN with Dropout (Adam)	1508.79
Deeper CNN with Dropout (SGD)	3091.07
Deeper CNN without Dropout	2949.31
MobileNetV2 Feature Extraction	1161.44
Fine-Tuned MobileNetV2	1148.62

Table X shows that the baseline CNN was the fastest model, while the deeper CNN with SGD took the longest time. The deeper CNN models needed more training time because they used more layers. MobileNetV2 used larger 224 * 224 images, but its training time was still lower than the SGD and no-dropout deeper CNN models. Overall, there was a trade-off between model performance and computational cost.

Adam vs SGD Optimizer Comparison

The deeper CNN was trained with both Adam and SGD to compare the effect of the optimizer. The Adam model achieved a validation accuracy of 0.9843 and a weighted F1-score of 0.9843. The SGD model performed better, with a validation accuracy of 0.9905 and a weighted F1-score of 0.9905.

However, SGD required much more training time. Adam took 1508.79 seconds, while SGD took 3091.07 seconds. This shows a clear trade-off. Adam was faster and more efficient, but SGD gave better final validation performance in this experiment.

	Optimizer	Accuracy	Precision	Recall	F1-score	Training_Time_Seconds
0	Adam	0.9843	0.9844	0.9843	0.9843	1508.7936
1	SGD	0.9905	0.9907	0.9905	0.9905	3091.0731

Figure 5: Validation performance comparison between Adam and SGD optimizers.

Figure 4: shows that Adam reached high validation accuracy earlier, while SGD improved more gradually. Although SGD took longer to train, it achieved the better final validation result.

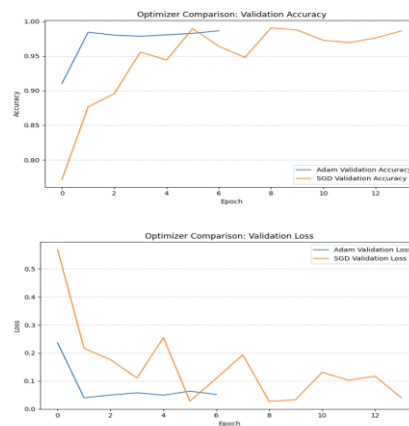


Figure 6: Validation accuracy and loss curves for Adam and SGD optimizers.

The validation curves show that Adam reached high validation accuracy earlier, while SGD improved more gradually. However, SGD achieved the stronger final validation result. The validation loss curve also shows that SGD ended with a lower loss, although it fluctuated more during training. This means Adam was faster and more stable early on, but SGD gave better final performance in this experiment.

Dropout Ablation and Overfitting Check

A dropout ablation study was carried out to test whether dropout improved the deeper CNN. The deeper CNN with dropout achieved a validation accuracy of 0.9843 and a weighted F1-score of 0.9843. After removing dropout, the model achieved a higher validation accuracy of 0.9992 and a weighted F1-score of 0.9992.

This result suggests that dropout reduced performance in this experiment. One possible reason is that the traffic sign classes had clear visual patterns, so removing dropout allowed the model to learn these patterns more strongly. However, because the validation score was very high, there was a possible risk of overfitting. For this reason, the model was later evaluated on the held-out test set to check generalization.

Validation performance comparison for dropout ablation:

Model	Accuracy	Precision	Recall	F1-score	Training Time(s)
Deeper CNN with Dropout	0.9843	0.9844	0.9843	0.9843	1508.79
Deeper CNN without Dropout	0.9992	0.9992	0.9992	0.9992	2949.31

Table X shows that the no-dropout model achieved higher validation scores than the dropout model, but it also required more training time. These supports selecting the no-dropout model for further evaluation, while still checking it on the held-out test set because the validation score was very high.

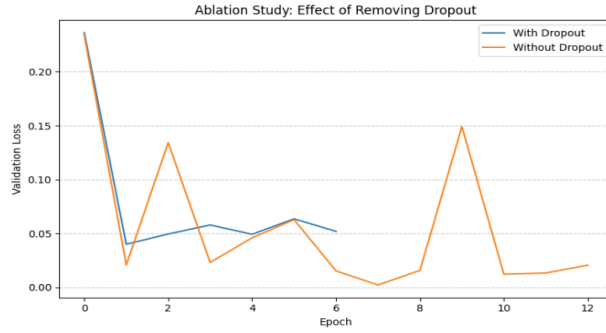


Figure 7: Validation loss comparison for the dropout ablation study.

The validation loss curve shows that the no-dropout model reached lower validation loss overall, but it also had some fluctuations. Because the no-dropout model achieved a very high validation score, the held-out test set was used later to check whether the model was overfitting or still generalizing well.

5. Fine-Tuning or Transfer Learning

MobileNetV2 was selected as the pre-trained model for transfer learning. It was loaded with ImageNet weights, so it had already learned general image features before being applied to the traffic sign dataset. For this experiment, the input image size was changed to 224×224 .

Two transfer learning approaches were tested. First, the MobileNetV2 convolutional base was frozen and used as a feature extractor. Only the new classification layers were trained for the five traffic sign classes. This model achieved a validation accuracy of 0.9183 and a weighted F1-score of 0.9185.

Second, the upper layers of MobileNetV2 were unfrozen and fine-tuned using a low learning rate. Batch normalization layers were kept frozen to make training more stable. Fine-tuning improved the validation accuracy to 0.9641 and the weighted F1-score to 0.9641. This shows that fine-tuning helped the pre-trained model adapt better to the traffic sign dataset.

However, the fine-tuned MobileNetV2 still performed below the best custom CNN models. This may be because MobileNetV2 was pre-trained on general ImageNet images, while this dataset contains more specific traffic sign patterns.

MobileNetV2 Feature Extraction	0.9183	0.9197	0.9183	0.9185
Fine-Tuned MobileNetV2	0.9641	0.9645	0.9641	0.9641

Figure 8: : Validation performance comparison between MobileNetV2 feature extraction and fine-tuning.

Overall Validation Performance Comparison

The overall validation comparison was used to select the final model. Among all tested models, the deeper CNN without dropout achieved the highest validation accuracy and weighted F1-score, both 0.9992. The SGD version of the deeper CNN also performed strongly with an accuracy of 0.9905, but it required the longest training time.

The MobileNetV2 models performed lower than the custom CNN models, although fine-tuning improved performance compared with frozen feature extraction. Overall, the validation results showed that the custom CNN models were more suitable for this dataset. Based on validation

performance, the deeper CNN without dropout was selected for final testing on the held-out test set.

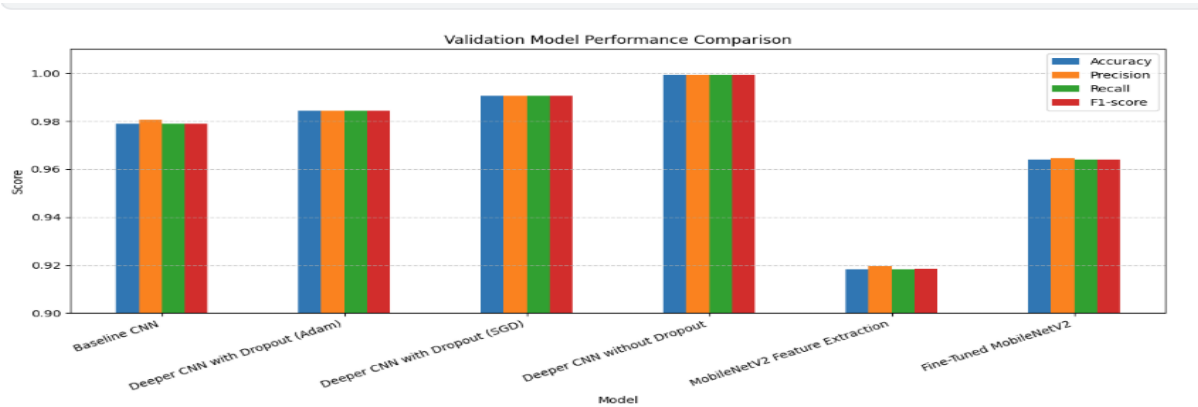


Figure 9: Overall validation performance comparison of all models.

The figure shows that the deeper CNN without dropout achieved the highest validation performance across accuracy, precision, recall, and F1-score. The SGD deeper CNN also performed strongly, but it required more training time. The MobileNetV2 models performed lower than the custom CNN models, although fine-tuning improved performance compared with frozen feature extraction.

Final Test Evaluation of Selected Model

The deeper CNN without dropout was selected for final testing because it achieved the best validation performance. The held-out test set was not used during model selection, so it was used only once to check how well the selected model generalised to unseen images.

On the held-out test set, the selected model achieved a test accuracy of 0.9864 and a weighted F1-score of 0.9864. This result is slightly lower than the validation score, which is expected because the test set was unseen during model selection. The model still generalised well overall.

	Selected_Model	Test_Accuracy	Test_Precision	Test_Recall	\
0	Deeper CNN without Dropout	0.9864	0.9868	0.9864	

	Test_F1-score
0	0.9864

Figure 10: Final test performance of the selected deeper CNN without dropout.

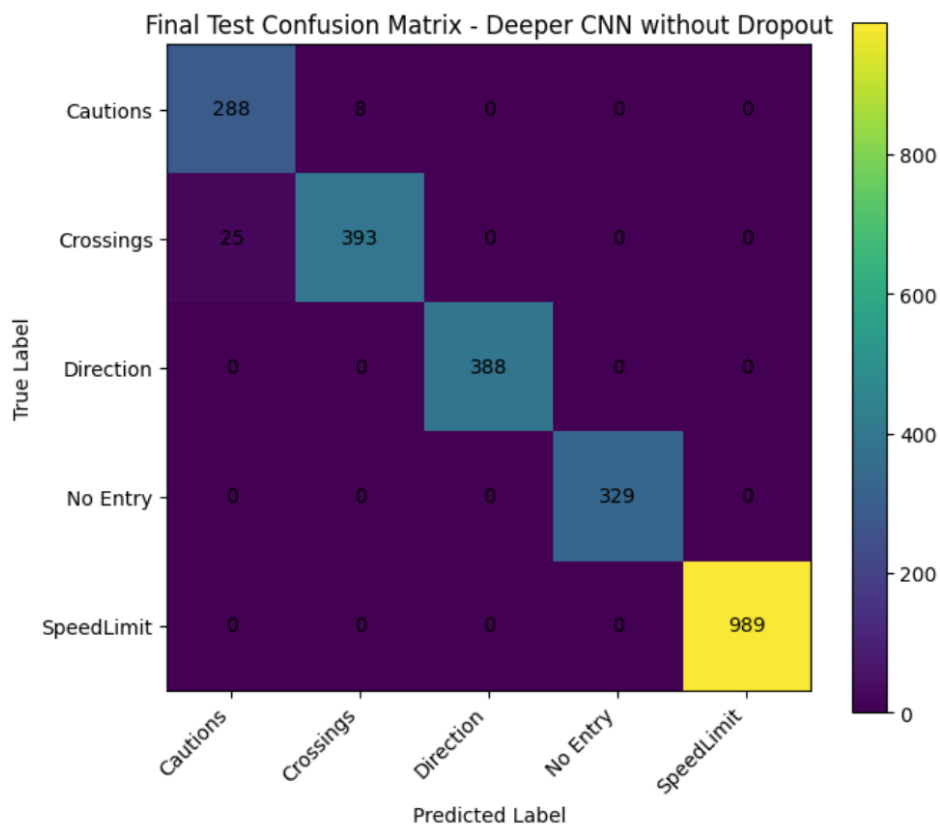


Figure 11: Final test confusion matrix for the selected deeper CNN without dropout.

The confusion matrix shows that Direction, No Entry, and SpeedLimit were classified very strongly. Most remaining errors were between Cautions and Crossings. This suggests that these two classes were harder to separate, possibly because they had fewer samples or more similar visual features.

Discussion

The results show that the custom CNN models performed better than the MobileNetV2 transfer learning models for this dataset. The baseline CNN already achieved strong performance, which suggests that the traffic sign classes had clear visual features. The deeper CNN improved on the baseline because the extra convolutional layers helped the model learn more detailed image patterns.

The optimizer comparison showed a trade-off between performance and training time. Adam trained faster, but SGD achieved better validation performance. This suggests that SGD was slower but gave a stronger result for the deeper CNN in this experiment.

The dropout ablation result was also important. The model without dropout achieved the best validation performance. This suggests that dropout may have reduced the model's ability to learn useful features from this dataset. However, because the validation score was very high, the final test evaluation was needed to check whether the model generalised well.

The final test result showed that the selected deeper CNN without dropout still performed strongly on unseen data, with 0.9864 test accuracy and 0.9864 weighted F1-score. The confusion matrix showed that Direction, No Entry, and SpeedLimit were classified very well. Most remaining mistakes were in Cautions and Crossings, which may have more similar visual features.

A limitation of this work is that the model was evaluated on a split created from the labelled training folder, rather than a fully independent external dataset. Also, the dataset was imbalanced,

with SpeedLimit having more images than the other classes. Future work could test the model on more real-world road images, use more balanced data, and try additional architecture or hyperparameter tuning.

Conclusion and Future Work

This project developed and compared several deep learning models for traffic sign classification. The experiments included baseline CNN, deeper CNN models, optimizer comparison, dropout ablation, and MobileNetV2 transfer learning. The deeper CNN without dropout achieved the best validation performance and was selected for final testing.

On the held-out test set, the selected model achieved 0.9864 accuracy and 0.9864 weighted F1-score. This shows that the model generalised well to unseen images. The results also showed that the custom CNN models performed better than the MobileNetV2 transfer learning models for this dataset. Fine-tuning improved MobileNetV2 compared with frozen feature extraction, but it remained below the best custom CNN model.

The main limitation is that the final test set was created from the cleaned labelled training data, rather than from a completely independent external dataset. The dataset was also imbalanced, with more SpeedLimit images than some other classes. In future work, the model could be tested on a larger and more balanced dataset with more real-world road conditions. Further improvements could also include stronger augmentation, hyperparameter tuning, and testing other pre-trained models such as ResNet or EfficientNet.